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Life-history strategy mediates the effects of multiple developmental stressors on Australian lizards

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Maternal stress and the developmental environment jointly affect offspring fitness and phenotype during early life. Mothers transmit stress to offspring through the deposition of hormones and energy into developing embryos, who are themselves subject to environmental stressors. A species' life-history strategy (e.g. size and number of eggs) may further buffer or exacerbate the effects of developmental stress on offspring. We experimentally tested the joint effects of stress hormones (exogenous corticosterone applied directly to eggs) and the developmental environment (incubation temperature) on the hatching success and size of two lizards (*Lampropholis delicata* and *Lampropholis guichenoti*) that differ in maternal investment strategy. Hatching success was reduced by corticosterone exposure, but only in *L. delicata*, which produces clutches of more numerous but smaller eggs than *L. guichenoti*. Hatchling size in both species, however, was reduced by corticosterone exposure, incubation at high temperatures or both. In both species, maternal investment (egg size) was the best predictor of hatching success and hatchling size, regardless of treatment. Our findings suggest that, both within species and between species, greater maternal investment buffers offspring from maternal and developmental stress effects on early-life fitness (survival) but does not prevent their effects on fitness-associated phenotypes (hatchling size).

1. Introduction

Global climate change increasingly exposes organisms to novel environmental stressors [1,2]. Beyond elevated temperatures, global climate change is predicted to affect organismal physiology and fitness through changes in water availability, salinity or pH, exposure to pollutants or diseases, or predation pressure and competition, which can act together additively, antagonistically or synergistically [3,4]. Consequently, multiple stressors and how they interact are now recognized as key to understanding how novel environments will impact populations ([5] and citations therein, reviewed in [6,7], freshwater ecosystems [8], marine ecosystems [3] and amphibians [9]). The adverse fitness effects of multiple stressors will have important implications for the persistence of populations, particularly at critical life stages [10–12]. Early life-history stages (e.g. embryonic, hatchling) are critically sensitive periods, where numerous traits like body plan and size are established, often with lifelong consequences for fitness [13]. Indeed, meta-analyses suggest that

the impacts of multiple stressors are more severe when experienced during embryonic life stages (e.g. [4]). However, most multi-stress work considers only the effects of simultaneously occurring stressors that act directly on a single individual, which overlooks how stressors acting on one generation may interact with those acting on another.

Stressors experienced by parental generations can result in dramatic physiological changes that affect offspring. Carry-over effects from parental generations experiencing the same or different stressors result in ‘legacy’ effects that can impact offspring in complex ways (i.e. parental/transgenerational effects [14–16]). It is essential to consider how intergenerational stressors act together with offspring environmental stress, particularly for egg-laying organisms, for several reasons. First, the timing of multiple stressors is more likely to coincide at early life stages because parents, particularly mothers, establish the developmental environment for their offspring. Second, early embryonic stages are highly sensitive to environmental perturbations [17,18], and there is limited opportunity for embryos of egg-laying organisms to mitigate stress through behavioural plasticity ([19]; but see [20]). Finally, understanding how stressors interact or not in early life stages is crucial for predicting individual survival and population growth [21–24]. Despite the potential importance of intergenerational stressors in influencing offspring stress responses, there is a paucity of studies applying a multi-stressor framework to this problem.

The mechanisms by which parents (specifically, mothers) transmit stress to their offspring are diverse, ranging from shifts in resource investment strategies [25], changes in maternal nesting behaviour [26,27], inheritance of epigenetic marks [28,29] or the deposition of hormones that have direct effects on embryonic development (reviewed in [30]; e.g. [31–33]). In vertebrates, one physiological response to stressors involves activation of the hypothalamic-pituitary-adrenal (HPA) axis that results in the release of glucocorticoid (stress) hormones [34,35]. Glucocorticoids induce physiological changes that help vertebrates cope with stressors but can have cascading effects on traits like reproductive investment and maternal behaviour, which in turn influence offspring size and survival [29,36]. Additionally, glucocorticoids can be directly transferred from mothers to offspring through deposition in yolk, which in turn influences offspring physiology and behaviour. Thus, offspring survival and phenotypes are a cumulative result of the complex three-way interactions between the maternal environment, the maternal phenotype and the offspring environment. Consequently, a major challenge in predicting how maternal and offspring stressors interact is that stress experienced by mothers results in a multitude of physiological, behavioural and epigenetic changes to the maternal phenotype that affect offspring and their environment both directly and indirectly (reviewed in [37–39]).

Lizards are excellent model systems for testing the effects of developmental stressors (e.g. glucocorticoids, incubation temperature) on offspring phenotype and fitness because embryonic development occurs almost entirely external to the mother. Using two congeneric skinks (*Lampropholis delicata* [40] and *Lampropholis guichenoti* [41]) with different life-history strategies (r- and K-selected, e.g. [42]), we tested how prenatal stress (simulated through the application of glucocorticoids to eggs) and developmental thermal stress interact to affect fitness-associated traits like egg mortality and the size at hatching. We also explore how these developmental conditions are mediated by maternal resource investment (i.e. egg mass). We predicted that prenatal corticosterone (CORT) exposure and warm incubation temperatures would independently increase embryonic mortality and reduce body size. Additionally, we predicted that prenatal CORT (the primary glucocorticoid in reptiles) and warm incubation temperatures would have additive effects, thereby further increasing embryonic mortality and reducing body size at hatching as the sum of their independent effects. Maternal investment, both within and between species, should act as a buffer to the stressors, such that larger eggs should be more robust to the effects of developmental stress on embryonic mortality and hatchling size. As such, we predicted that smaller eggs within species would be more affected by developmental conditions than larger eggs, and *L. guichenoti* eggs would be more robust than *L. delicata*.

2. Methods

(a) Study system and animal husbandry

We used two congeneric lizards—the delicate skink (*L. delicata*) and the garden skink (*L. guichenoti*)—to test the hypothesis that ‘maternally derived’ stress, the developmental environment and maternal investment jointly influence offspring survival and phenotypes. Both species are common in semi-urban habitats (e.g. leaf litter, gardens) throughout their widespread distribution in southeastern Australia, where they occupy near-identical ecological niches as small-bodied (approximately 1–2 g adult size), relatively short-lived (approximately 2–3 years), active foragers of small invertebrates [43,44]. However, these two species diverge in their reproductive investment strategy [45]. *Lampropholis delicata* lays more numerous (2–6 eggs), smaller (approximately 0.145 g) eggs per clutch (more r-selected), while *L. guichenoti* lays fewer (1–4 eggs per clutch), larger (approximately 0.188 g) eggs per clutch (more K-selected). By using two ecologically identical, congeneric lizards with divergent reproductive investment strategies as a model system, we aimed to assess how the joint effects of (simulated) maternal and developmental stress (see §2b) on offspring development are buffered or exacerbated by variation in maternal investment both within and between species.

All eggs used in this study were from a breeding colony established from wild populations of *L. delicata* and *L. guichenoti* caught in 2019 in Sydney, Australia. The parent generation from which we collected eggs in 2022/2023, therefore, primarily consisted of first-generation lizards reared entirely from oviposition under laboratory conditions. Lizards were housed in communal enclosures (41.5 L × 30.5 W × 21 H cm) containing two males and four females of a single species. Lizards were fed three times per week with approximately 40 crickets (*Acheta domesticus*) dusted with reptile calcium and multivitamin supplements and provided water daily. Enclosures were fitted with a cabinet liner, polyvinyl chloride (PVC) pipe refuges and eucalyptus bark. A 22–32°C temperature gradient was generated in each enclosure using heat cords, and enclosures were lit with UVA/UVB bulbs on a 12 h light : dark cycle. Throughout the reproductive season (October 2022–February 2023), a nesting

box (12.5 L × 8.3 W × 5 H cm) filled with moist vermiculite was placed on the warm side of the thermal gradient in each enclosure. We checked nesting boxes for eggs three times per week.

(b) Treatment and data collection

Upon finding eggs, we measured the length and width with digital callipers (± 0.01 mm) and the mass (± 0.001 g) on a digital scale (OHAUS, Model spx123). We pseudo-randomly assigned eggs from each clutch to each of four 'maternal stress' (CORT or control) by incubation temperature (23°C or 28°C) treatments, such that we attempted to balance each clutch across treatments ('split-clutch' design; figure 1). For the 'maternal stress' treatment, 5 μ l of either CORT dissolved in 100% ethanol (10 pg CORT ml⁻¹) or a 100% ethanol only control was applied topically through a pipette to the side of each egg containing the yolk. When topically applied to *L. delicata* eggs, this concentration and amount of CORT solution increases circulating CORT concentrations in the yolk to 2 s.d. above the naturally occurring mean and is within the natural range of CORT variation [29]. Following stress treatment, eggs were placed in 80 ml cups containing moist vermiculite (12 g water to 4 g vermiculite) and covered with plastic wrap, then placed in one of two incubators (LATWIT 2X5D-R1160) programmed to either a 'cool' (23 $\pm$ 3°C) or 'warm' (28 $\pm$ 3°C) thermal regime. These thermal regimes represent the lower and upper ends of the natural range of incubation temperatures for these two species [46,47]. Specifically, 23°C approximates the average temperature of shaded nests in natural populations, while 28°C approximates the average temperature of unshaded, heat-exposed nests in open grasslands [48,49]. Both temperatures are also approximately equal in magnitude to the optimal temperature for hatching success (26°C; at least in *L. delicata*) and within the range of optimal temperatures for development for these species [50]. We checked the incubators three times per week for hatchlings and dead eggs. We determined that eggs were dead if desiccated or covered in mould. We recorded the date dead eggs were found and disposed of them. Upon finding hatchlings, we recorded the hatch date, then measured their mass (± 0.001 g) on a digital scale (OHAUS, Model spx123) and their snout-vent length (SVL; ± 1.0 mm) and tail length (± 1.0 mm) using a ruler.

(c) Statistical analyses

All processing and analyses were performed in R v. 4.2.1 [51]. All data and code are available as a public GitHub repository [52]. Before hypothesis testing, we performed a model selection process to simplify models, but always retained our hypothesized main effects. We constructed separate Bayesian generalized linear mixed-effects models (GLMMs) for egg mortality and hatchling mass for each species. Comparison models included each permutation of the additive, two-way and three-way interactions between stress treatment (control or CORT), incubation temperature (23°C or 28°C) and egg mass as fixed effects terms. Species were not included as fixed effects terms in comparison or analysis models. Instead, we modelled the egg mortality and hatchling mass separately for each species to reduce model complexity and because the Bayesian framework allows for comparisons of fixed effects (e.g. CORT, temperature) between species even if species is not included as a term in models. Clutch identity was included as a random intercept in all models to control for the effects of eggs coming from the same mother. Because egg mortality was a binomial response variable (1 if they died, 0 if they hatched), it was modelled assuming a Bernoulli error distribution ('logit' link function).

Hatchling mass is a numeric trait that was modelled assuming a Gaussian error distribution. We fitted all models for comparison in rstan (v. 2.32.6; [53]) using the brms (v. 2.18.0; [54–56]) R package. All comparison models were run on four parallel Markov chain Monte-Carlo (MCMC) sampling chains for 2000 iterations with 1000 warm-up iterations at an adapt delta value of 0.99. We compared models using the 'loo' package in R (v. 2.8.0; [57]) by first fitting expected logwise predicted densities (ELPD) to each model, then comparing the difference in ELPD between each model ('loo_compare' function). The ELPD estimates the predictive accuracy of each model using leave-one-out (LOO) cross-validation and returns a score of model fit which can be compared amongst candidate models, similar to Aikake information criterion (AIC) [58]. We determined the model of best fit as the model with the highest ELPD value among the models being compared. If the ELPD values and their error overlapped between multiple models, we considered these models statistically indistinguishable from each other. If multiple models were indistinguishable, we defaulted to the simplest model with the fewest interaction terms among those models.

For both egg mortality and hatchling mass, the best-fitting models for both species included the additive effects of stress treatment, incubation temperature treatment and starting egg mass as fixed effects with no interaction terms, and clutch identity was included as a random intercept. When the best-fitting model structures were determined, we refitted the best models, running them on four MCMC chains for 5000 iterations with 2500 warm-up iterations at an adapt delta value of 0.99. We inspected summaries of the model outputs for chain convergence (Rhat < 1.02) and sufficient effective and bulk sample sizes (both > 1000) before further analyses.

For hypothesis testing, we tested the independent effects of temperature, CORT treatment and their interaction by constructing relevant posterior distributions for comparisons of interest and evaluating significance using the pMCMC value (pMCMC < 0.05). The pMCMC value tests the hypothesis that the difference between predicted posterior distributions differs from zero.

We also tested the independent effect of egg mass on embryonic mortality, because we predicted that egg mass, regardless of treatment, would be the greatest predictor of embryonic mortality. Again, to compare the effects of CORT and temperature between species, we extracted the posterior distributions generated for each main effect (i.e. stress = CORT – control; temperature = 28–23°C) for each species. Then, we subtracted the 'effect distribution' of *L. guichenoti* from that of *L. delicata* (i.e. stress = stress_{delicata} – stress_{guichenoti}; temperature = temperature_{delicata} – temperature_{guichenoti}) to make a distribution of differences

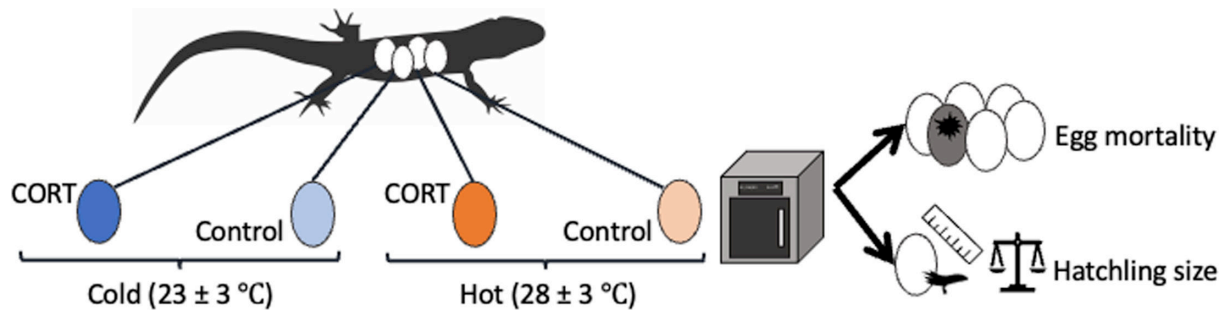


Figure 1. Schematic of experimental design. Eggs from each clutch of lizards (*L. delicata* and *L. guichenoti*) were divided evenly among four ‘maternal stress’ (CORT or control) by incubation temperature (23°C or 28°C) treatments using a 2 × 2 factorial split-clutch design. Treated eggs were incubated until they either died or hatched. Upon hatching, we measured the mass and SVL of hatchlings.

in effect sizes. From the resulting species comparison distribution, we tested for differences in effects between species using the pMCMC value of the distribution.

As an indicator of effect sizes, we also estimated the mean (estimate) and s.e. (estimate error) of each resulting distribution. For comparisons of treatment levels, species or effect sizes, the estimate and estimate error are reported as the mean and s.e. of the difference between distributions (denoted with Δ). For the main effect of egg size, or for individual treatment levels, the estimate and estimate error are the mean and s.e. of the predicted posterior distribution generated by the model (denoted with β).

3. Results

We treated a total of 684 *L. delicata* eggs from 160 unique clutches sourced from 44 communal enclosures and 568 *L. guichenoti* eggs from 197 unique clutches from 30 communal enclosures. Due to the split-clutch design we used, the eggs from each clutch were roughly evenly divided among each of the four temperature-by-CORT treatment combinations. For *L. delicata*, 168 eggs from 140 clutches were assigned to the ‘control, 23°C’ treatment, 172 eggs from 141 clutches to ‘control, 28°C’, 168 eggs from 144 clutches to ‘CORT, 23°C’ and 176 eggs from 144 clutches to ‘CORT, 28°C’. For *L. guichenoti*, 143 eggs from 136 clutches were assigned to the ‘control, 23°C’ treatment, 140 eggs from 133 clutches to ‘control, 28°C’, 138 eggs from 134 clutches to ‘CORT, 23°C’ and 147 eggs from 142 clutches to ‘CORT, 28°C’.

(a) Do maternal and developmental stress jointly influence survival?

Stress treatment affected *L. delicata* egg mortality, with CORT-treated eggs more likely to die than control eggs (Δ CORT – control; estimate (Est.) $\pm$ estimate error (Est. Err.) = 0.713 ± 0.300 ; pMCMC = 0.015). However, there was no effect of CORT on the survival of *L. guichenoti* eggs (Δ CORT – control; Est. $\pm$ Est. Err. = 0.466 ± 0.397 ; pMCMC = 0.229). Incubation temperature did not affect the egg mortality of *L. delicata* (Δ 28–23°C; Est. $\pm$ Est. Err. = 0.71 ± 0.295 ; pMCMC = 0.208) nor *L. guichenoti* (Δ 28–23°C; Est. $\pm$ Est. Err. = 0.218 ± 0.397 ; pMCMC = 0.579). Egg mass alone affected the mortality of *L. guichenoti*, such that smaller eggs were more likely to die than larger eggs (β egg mass; Est. $\pm$ Est. Err. = -19.015 ± 7.483 ; pMCMC = 0.009). Egg mass similarly influenced the mortality of *L. delicata*, but this effect was non-significant (β egg mass; Est. $\pm$ Est. Err. = -15.350 ± 8.265 ; pMCMC = 0.059). Posterior distributions derived from Bayesian GLMMs of embryonic mortality and proportions of hatching success from raw data are provided in figure 2. Summary statistics on embryonic mortality are provided in table 1. Model outputs of CORT, incubation temperature and maternal investment effects on embryonic mortality are provided in table 2.

(b) Does developmental stress affect fitness-related phenotypes?

Hatchling mass was independently affected by stress treatment, incubation temperature and starting egg mass in both *L. delicata* and *L. guichenoti* (all pMCMCs < 0.001). Starting egg mass was the strongest predictor of hatchling mass in both *L. delicata* (β egg mass; Est. $\pm$ Est. Err. = 0.310 ± 0.022) and *L. guichenoti* (β egg mass; Est. $\pm$ Est. Err. = 0.296 ± 0.021). In both species, larger eggs yielded larger hatchlings, regardless of treatment group. Treatment with CORT affected hatchling mass in both *L. delicata* (Δ CORT – control; Est. $\pm$ Est. Err. = -0.003 ± 0.001) and *L. guichenoti* (Δ CORT – control; Est. $\pm$ Est. Err. = -0.003 ± 0.001), such that CORT-treated eggs yielded smaller hatchlings than control eggs. Incubation temperature similarly affected hatchling mass in both *L. delicata* (Δ 28 – 23°C; Est. $\pm$ Est. Err. = -0.004 ± 0.001 ; pMCMC < 0.001) and *L. guichenoti* (Δ 28 – 23°C; Est. $\pm$ Est. Err. = -0.003 ± 0.001 ; pMCMC = 0.001), with eggs incubated at 28°C yielding smaller hatchlings than eggs incubated at 23°C. The effects of CORT treatment and incubation temperature on hatchling mass were additive in both species. Compared with control eggs incubated at 23°C (*L. delicata*: $N_{\text{control},23^\circ\text{C}} = 152$, mass = 0.112 ± 0.013 g; *L. guichenoti*: $N_{\text{control},23^\circ\text{C}} = 134$, mass = 0.142 ± 0.016 g), treatment with CORT (*L. delicata*: $N_{\text{CORT},23^\circ\text{C}} = 145$, mass = 0.108 ± 0.013 g; *L. guichenoti*: $N_{\text{CORT},23^\circ\text{C}} = 127$, mass = 0.140 ± 0.015 g) or incubation at 28°C (*L. delicata*: $N_{\text{control},28^\circ\text{C}} = 149$, mass = 0.106 ± 0.012 g; *L. guichenoti*: $N_{\text{control},28^\circ\text{C}} = 131$, mass = 0.140 ± 0.017 g) resulted in smaller hatchlings. Simultaneous treatment with CORT and incubation at 28°C resulted in the smallest hatchlings among treatment groups (*L. delicata*: $N_{\text{CORT},28^\circ\text{C}} = 141$, mass = 0.105 ± 0.012 g; *L. guichenoti*: $N_{\text{CORT},28^\circ\text{C}} = 132$, mass = 0.137 ± 0.017 g). Posterior distributions derived from Bayesian GLMMs of embryonic mortality and stat summary plots of

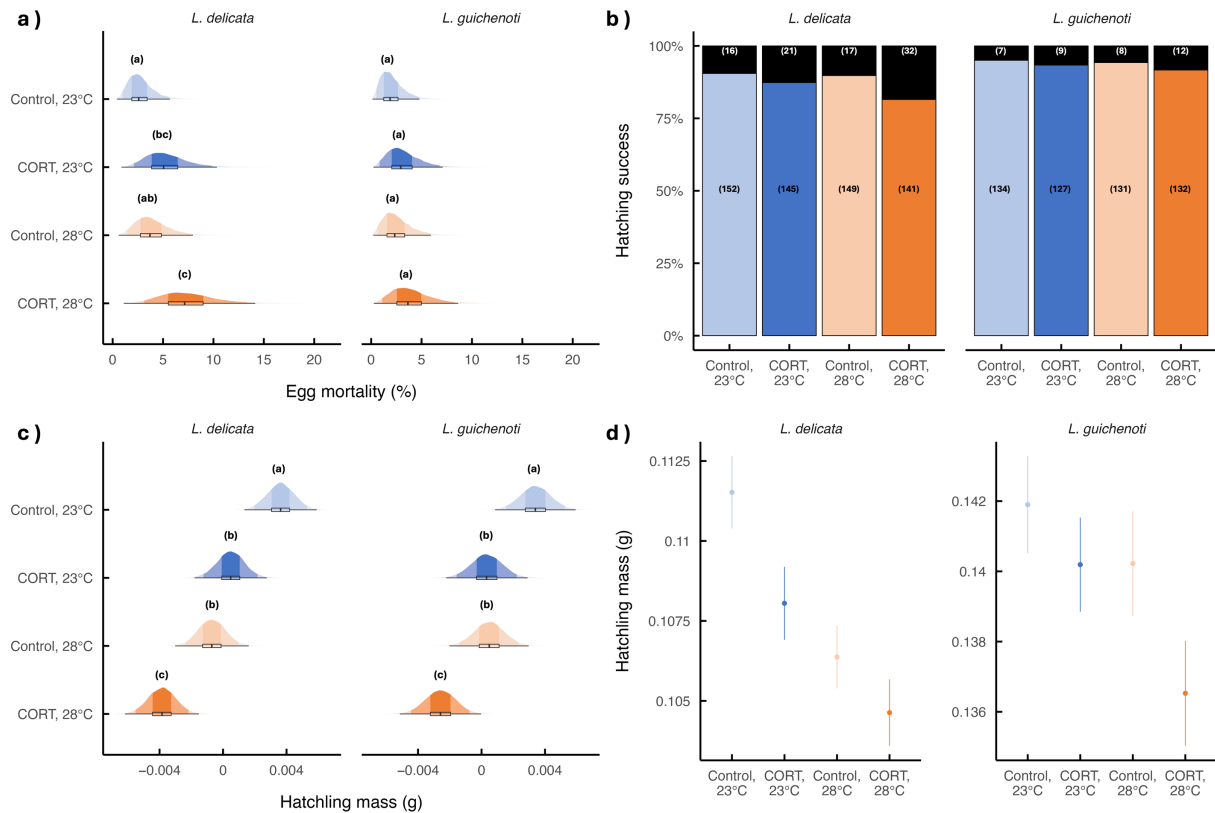


Figure 2. Effects of prenatal stress (CORT or control) and incubation temperature (23°C or 28°C) on embryonic mortality and hatchling mass in *L. delicata* and *L. guichenoti*. Plots (a) and (c) are density plots overlaid with box-and-whisker plots derived from the predicted posterior distributions of each treatment level generated by Bayesian GLMM of embryonic mortality and hatchling mass, respectively. Shaded regions represent the 50%, 80% and 95% confidence intervals of each distribution, while the centre of the box-and-whisker plot is the mean. Letters in brackets indicate similarities among treatment levels within each species. Embryonic mortality is the percentage of mortality explained by the treatment level. Plot (b) is a bar chart derived from raw data. Coloured regions are the proportion of eggs that survived to hatching in each treatment level within each species. Black regions are the proportion of eggs that died. Numbers in brackets are the total number of eggs that survived or died. Plot (d) is a summary statistic plot derived from raw data. Dots are the mean hatchling mass in each treatment group and lines are the standard error.

Table 1. Summary statistics of the raw data for each treatment group within each species. Eggs (died) and eggs (hatched) are counts of eggs that either died or hatched, respectively. Egg mass (died) and egg mass (hatched) are the initial mass (in grams) of eggs that either died or hatched, respectively. Hatchling mass is the size of hatchlings from eggs that successfully hatched. All mass values are given as mean $\pm$ s.d.

species	trait	control, 23°C	CORT, 23°C	control, 28°C	CORT, 28°C
<i>L. delicata</i>	eggs (died)	16	21	17	32
<i>L. delicata</i>	eggs (hatched)	152	145	149	141
<i>L. delicata</i>	egg mortality	9.50%	12.70%	10.20%	18.50%
<i>L. delicata</i>	egg mass (died)	0.126 $\pm$ 0.032	0.138 $\pm$ 0.024	0.142 $\pm$ 0.039	0.137 $\pm$ 0.03
<i>L. delicata</i>	egg mass (hatched)	0.145 $\pm$ 0.027	0.146 $\pm$ 0.026	0.146 $\pm$ 0.024	0.147 $\pm$ 0.022
<i>L. delicata</i>	hatchling mass (g)	0.112 $\pm$ 0.013	0.108 $\pm$ 0.013	0.106 $\pm$ 0.012	0.105 $\pm$ 0.012
<i>L. guichenoti</i>	eggs (died)	7	9	8	12
<i>L. guichenoti</i>	eggs (hatched)	134	127	131	132
<i>L. guichenoti</i>	egg mortality	5%	6.60%	5.80%	8.30%
<i>L. guichenoti</i>	egg mass (died)	0.132 $\pm$ 0.026	0.182 $\pm$ 0.059	0.176 $\pm$ 0.046	0.169 $\pm$ 0.043
<i>L. guichenoti</i>	egg mass (hatched)	0.184 $\pm$ 0.027	0.186 $\pm$ 0.029	0.185 $\pm$ 0.029	0.188 $\pm$ 0.035
<i>L. guichenoti</i>	hatchling mass (g)	0.142 $\pm$ 0.016	0.14 $\pm$ 0.015	0.14 $\pm$ 0.017	0.137 $\pm$ 0.017

hatchling mass from raw data are provided in figure 2. Summary statistics on hatchling mass are provided in table 1. Model outputs of CORT, incubation temperature and maternal investment effects on hatchling mass are provided in table 2.

Table 2. Summary of the main effects of prenatal stress (CORT or control), incubation temperature (23°C or 28°C) and maternal investment (egg mass) on embryonic mortality and hatchling size of *L. delicata* and *L. guichenoti*. All values are derived from Bayesian GLMM. For each response variable and each species, the independent effects of stress (CORT – control), temperature (28–23°C) and egg mass are summarized. The estimate is the mean of the difference between predicted posterior distributions (stress and temperature), estimate error is the error about the resultant distribution and l-95% CI and u-95% CI are the lower and upper 95% confidence intervals of the distribution, respectively. The pMCMC value is a test statistic suitable for testing whether two or more predicted posterior distributions overlap, with pMCMC < 0.05 indicating that the difference between distributions is not equal to zero.

trait	species	contrast	estimate	estimate error	l-95% CI	u-95% CI	pMCMC
egg mortality	<i>L. delicata</i>	intercept	−3.651	0.456	−4.615	−2.83	≤ 0.001
egg mortality	<i>L. delicata</i>	stress (CORT – control)	0.696	0.309	0.109	1.328	0.02
egg mortality	<i>L. delicata</i>	temperature (28–23°C)	0.366	0.3	−0.228	0.946	0.22
egg mortality	<i>L. delicata</i>	egg mass	−16.062	8.457	−32.985	0.637	0.057
egg mortality	<i>L. guichenoti</i>	intercept	−4.004	0.581	−5.267	−3	≤ 0.001
egg mortality	<i>L. guichenoti</i>	stress (CORT – control)	0.463	0.404	−0.323	1.273	0.25
egg mortality	<i>L. guichenoti</i>	temperature (28–23°C)	0.22	0.397	−0.561	1.004	0.578
egg mortality	<i>L. guichenoti</i>	egg mass	−19.081	7.645	−34.265	−4.326	0.011
hatchling mass	<i>L. delicata</i>	intercept	0.004	0.001	0.002	0.005	≤ 0.001
hatchling mass	<i>L. delicata</i>	stress (CORT – control)	−0.003	0.001	−0.004	−0.002	≤ 0.001
hatchling mass	<i>L. delicata</i>	temperature (28–23°C)	−0.004	0.001	−0.005	−0.003	≤ 0.001
hatchling mass	<i>L. delicata</i>	egg mass	0.31	0.022	0.267	0.353	≤ 0.001
hatchling mass	<i>L. guichenoti</i>	intercept	0.003	0.001	0.002	0.005	≤ 0.001
hatchling mass	<i>L. guichenoti</i>	stress (CORT – control)	−0.003	0.001	−0.005	−0.001	≤ 0.001
hatchling mass	<i>L. guichenoti</i>	temperature (28–23°C)	−0.003	0.001	−0.005	−0.001	0.001
hatchling mass	<i>L. guichenoti</i>	egg mass	0.297	0.02	0.257	0.337	≤ 0.001

(c) Does life-history strategy buffer offspring from developmental stress?

Despite CORT treatment having a direct effect on the embryonic mortality of *L. delicata*, but no direct effect on *L. guichenoti*, there were no detectable differences in the effects of CORT on mortality between species ($\Delta \text{stress}_{\text{delicata}} - \text{stress}_{\text{guichenoti}}$; Est. $\pm$ Est. Err. = 0.244 ± 0.044 ; pMCMC = 0.622). There were no direct effects of incubation temperature on the mortality of either *L. delicata* or *L. guichenoti*, and the effects of incubation temperature on mortality did not differ between species ($\Delta \text{temperature}_{\text{delicata}} - \text{temperature}_{\text{guichenoti}}$; Est. $\pm$ Est. Err. = 0.154 ± 0.003 ; pMCMC = 0.758). Similarly, neither the effects of CORT ($\Delta \text{stress}_{\text{delicata}} - \text{stress}_{\text{guichenoti}}$; Est. $\pm$ Est. Err. = 0.000 ± 0.000 ; pMCMC = 0.943) nor incubation temperature ($\Delta \text{temperature}_{\text{delicata}} - \text{temperature}_{\text{guichenoti}}$; Est. $\pm$ Est. Err. = -0.001 ± 0.000 ; pMCMC = 0.153) on hatchling mass differed between *L. delicata* and *L. guichenoti*. Results of between-species comparisons of the main effects of prenatal CORT and incubation temperature are provided in table 3.

4. Discussion

We found that prenatal CORT induced greater embryonic mortality compared with the control treatment, but only significantly in *L. delicata*. Rather, in both *L. delicata* and *L. guichenoti*, maternal investment (egg size) was the strongest predictor of survival to hatching, with ‘more stressed’ (CORT-treated or warm-incubated) eggs requiring greater maternal investment to hatch than ‘less stressed’ (control-treated or cool-incubated) eggs. In contrast, while egg size positively influenced hatchling size, treating eggs with CORT and incubating eggs at high temperatures resulted in smaller hatchlings of both *L. delicata* and *L. guichenoti*. Our findings suggest: (i) maternal stress (simulated through CORT application) induces embryonic mortality, but variation in incubation temperature does not, (ii) variation in maternal investment both within- and among-species buffers developing offspring from the lethal effects of developmental stressors and (iii) exposure to multiple developmental stressors affects post-embryonic fitness-related traits like body size regardless of maternal investment strategy.

(a) Maternally derived stress is lethal to embryos

Treatment with prenatal CORT alone increased embryonic mortality in *L. delicata*, while incubation temperature had weak or no influence on mortality in either *L. delicata* or *L. guichenoti*. This finding suggests that treating eggs with exogenous CORT, which mimics an increase in maternally derived stress hormones, challenges successful development, but warm incubation temperatures representing heat-exposed nests did not. The lack of thermal effects on hatching success is perhaps unsurprising,

Table 3. Comparison of the main effects between *L. delicata* and *L. guichenoti*. Using the predicted posterior distributions from Bayesian GLMMs, we calculated the distribution of each main effect (prenatal CORT or incubation temperature) on egg mortality and hatchling mass in each species (see table 2). Then, we compared the distributions of each resultant main effect distribution between species to test the hypothesis that prenatal CORT and incubation temperature have different effects on *L. delicata* and *L. guichenoti*. The estimate is the mean of the difference between predicted posterior distributions (*L. delicata* – *L. guichenoti*), estimate error is the error about the resultant distribution and I-95% CI and u-95% CI are the lower and upper 95% confidence intervals of the distribution, respectively. The pMCMC value is a test statistic suitable for testing whether two or more predicted posterior distributions overlap, with pMCMC < 0.05 indicating that the difference between distributions is not equal to zero.

trait	effect	contrast	estimate	estimate error	I-95% CI	u-95% CI	pMCMC
egg mortality	stress (CORT – control)	<i>L. delicata</i> – <i>L. guichenoti</i>	0.233	0.004	–0.771	1.242	0.643
egg mortality	temperature (28°–23°C)	<i>L. delicata</i> – <i>L. guichenoti</i>	0.146	0.004	–0.829	1.112	0.768
hatchling mass	stress (CORT – control)	<i>L. delicata</i> – <i>L. guichenoti</i>	≤ 0.001	≤ 0.001	–0.002	0.002	0.943
hatchling mass	temperature (28°–23°C)	<i>L. delicata</i> – <i>L. guichenoti</i>	–0.001	≤ 0.001	–0.003	0.001	0.153

as the incubation temperatures we used were designed to be similar to natural nest temperatures rather than thermal extremes, which indicates that lizard embryos are relatively robust to thermal stress across at least a narrow range of incubation regimes. However, CORT reduced hatching success in *L. delicata* but not *L. guichenoti*, raising the question of why hatching success is affected by maternally derived stress in some species but not others. We propose that glucocorticoids (e.g. CORT, cortisol) modulate the balance between maternal investment (energy availability) and the cost of development (energy demand).

As the primary stress hormone in reptiles, CORT makes energy available for mounting the energetically demanding physiological stress response [36,59,60]. Prenatal CORT is metabolized by embryos, thereby increasing baseline metabolic rates. As such, prenatal CORT functionally increases developmental costs and can rapidly consume resources (e.g. yolk) that would be otherwise available for successful development (e.g. [61]). Additionally, energy is made available for the stress response by shunting energy away from other biological processes (e.g. immune function), thereby potentiating embryonic sensitivity to lethal stressors (e.g. infection). Similarly, incubation at high temperatures is frequently associated with elevated metabolic rates, reduced yolk incorporation and greater total energy requirements for development [62,63]. Consequently, exposure to CORT, high incubation temperatures, or both can induce developmental costs to exceed the energy available for maintaining development and responding to additional challenges, resulting in hatching failure. If energetic mismatches underlie the effects of CORT or poor thermal environments on embryonic mortality, then it follows that species or individuals with more energy for development (i.e. those receiving greater maternal investment) would be more robust to developmental stress. Indeed, this is precisely the pattern we observed between and within species in the present study. Future studies should explore this possibility by investigating how CORT directly affects energy expenditure (e.g. metabolic rate) in eggs or through experiments manipulating both CORT and the amount of energy (yolk) available to eggs.

(b) Maternal investment buffers embryos from lethal stress effects

Consistent with general predictions of life-history theory [64], embryonic mortality was higher in the more r-selected (more numerous, smaller eggs) *L. delicata* than the more K-selected (fewer, larger eggs) *L. guichenoti*, regardless of CORT treatment or incubation temperature. Additionally, the greater per-offspring investment provided to *L. guichenoti* eggs relative to *L. delicata* eggs [65] may also have buffered the increased developmental costs associated with elevated CORT or high incubation temperatures, explaining why there was no effect of CORT on *L. guichenoti* mortality, but also no difference in the effect of CORT on mortality between species. In other words, the patterns of mortality associated with CORT were similar between species, just weaker in *L. guichenoti*. Furthermore, the strongest predictor of hatching success for both *L. delicata* and *L. guichenoti* was the natural variation in egg mass, which is representative of maternal investment. Within each species, larger eggs were more likely to survive than smaller eggs, regardless of CORT treatment or incubation temperature. Interestingly, however, the average size of hatched eggs versus those that died varied among treatment groups within each species. Eggs that died in the CORT-treated, hot-incubated and simultaneously CORT-treated and hot-incubated groups were larger on average than control eggs that died. However, the average size of the eggs that hatched was similar among all experimental groups. We interpret this as a threshold minimum amount of energy (i.e. yolk or egg mass) required for eggs to develop, and exposure to one or more physiological stressors (e.g. CORT, incubation temperature) raises that minimum. Our results suggest that mothers may improve the hatching success of their eggs under stressful environmental conditions by increasing resource investment. If stressful conditions persist across multiple generations, such as we would expect under current climate change projections, the ‘filtering’ of small eggs could induce more r-selected populations or species (like *L. delicata*) to shift towards K-selected life-history strategies that are more robust to developmental stress (like *L. guichenoti*).

Although increasing egg size may increase fitness in stressful environments, species may be limited in their ability to shift towards greater investment due to physical or physiological constraints on the size or number of eggs produced in each clutch [66–68]. Additionally, the stressful environments in which adopting a greater per-offspring investment strategy would benefit hatching success will likely limit the resources available to mothers for reproduction. Furthermore, producing or carrying larger eggs in such environments may impose fitness costs on the mother that outweigh the benefits of improving offspring survival or maximizing offspring size or quality (reviewed in [39,69]; e.g. [70–72]). Indeed, a more realistic scenario—relative to this

experiment where resources available to mothers for reproduction were virtually unlimited—is that not only are eggs subjected to multiple stressors like maternally derived CORT and high incubation temperatures, but also that the amount of energy available to eggs for coping with these stressors and completing development is reduced. Therefore, the interactions between maternally derived stress, thermal environments and maternal investment are likely to be a strong determinant of hatching success in natural environments, with potential consequences for fitness-associated traits post-hatching.

(c) Developmental stressors have lingering effects on fitness-related traits

Although *Lampropholis* embryonic mortality was relatively robust to the combined effects of CORT treatment and incubation temperature, and evidence suggests that both within- and between-species variation in reproductive investment strategy acts as a buffer to these effects, the same could not be said for hatchling size in either *L. delicata* or *L. guichenoti*. Unsurprisingly, larger eggs generally resulted in larger hatchlings regardless of CORT or incubation temperature in both species. However, treatment with CORT or high temperature alone resulted in smaller hatchlings, and treatment with both CORT and high temperature resulted in even smaller hatchlings relative to untreated control eggs incubated at cool temperatures in both species. Size at hatching is a strong determinant of survival to adulthood in many reptiles, as well as growth and adult size, which in turn influences reproductive output ([73,74]; e.g. [75–77]; but see [78]). Thus, while embryonic mortality may be relatively robust to multiple stressors like CORT and incubation temperature, these developmental stressors induce phenotypic effects that will likely persist into adulthood and the next generation.

We note, however, that the effects of CORT and high temperatures on hatchling body size were relatively small. CORT treatment or warm incubation induced an approximately 0.003 g reduction in body size on average, while treatment with both CORT and warm incubation temperatures resulted in an approximately 0.006 g reduction in hatchling size. Even so, these are small-bodied lizards, especially *L. delicata*, which hatched at 0.108 g on average. Therefore, the cumulative effects of CORT and warm incubation temperature resulted in an approximately 6% reduction in *L. delicata* body mass (approximately 3.5% for *L. guichenoti*). Although it seems unlikely these mild reductions in hatchling body mass will carry strong fitness consequences under laboratory conditions, even small reductions in body size may be significant in natural environments when environmental conditions are more extreme, resources are limited, competition is high or juveniles are under predation pressure (e.g. [79,80]). This may be especially true if reduced body size limits an individual's ability to acquire food that would allow them to compensate for their smaller size at hatching, thereby leading to a pattern of reduced body size or body condition throughout life compared with their larger-bodied peers [81–83].

Beyond their effects on early-life size, stressors experienced during development can have long-lasting mechanistic effects on multiple biological systems throughout an organism's life ([13]; reviewed in [29,84,85]). Indeed, developmental CORT exposure and incubation temperature can permanently alter immune function [86], HPA axis reactivity [87,88], growth and metabolism [89–93], ageing and lifespan [87,93], reproduction [94], cognition [95–97] and behaviour [98,99]. If stressors like increased temperature persist across multiple generations, as we would expect in the face of global climate change, even mild developmental challenges acting on one generation may have far-reaching consequences for the health, persistence and evolution of populations into future generations.

Ethics. All experimental procedures were approved by the Australian National University Animal Experimentation Ethics Committee (Protocol Number A2021/56).

Data accessibility. All data, code and supplementary materials are available as a public GitHub repository archived in Zenodo [52].

Supplementary material is available online [100].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. D.C.L.: conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing—original draft, writing—review and editing; P.R.: conceptualization, investigation, methodology, visualization, writing—original draft, writing—review and editing; A.Y.P.: investigation, methodology, writing—review and editing; O.L.C.: conceptualization, methodology, resources, validation, writing—review and editing; C.F.: conceptualization, funding acquisition, methodology, project administration, writing—review and editing; D.W.A.N.: conceptualization, funding acquisition, methodology, project administration, supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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